Visualizing Eye Movement Data in Dynamic Object Tracking Tasks

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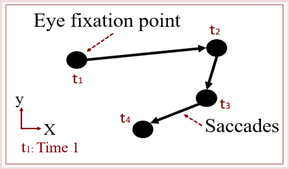
Introduction

- Analysis of **eye movement (EM)** data helps to understand the **cognitive process** during visual search tasks.
- **Visualization** of the EM data - first step in this analysis



- EM data = **time-ordered sequence** of 2-D coordinate locations
- **EM data has two parts**
  - Fixation point : (x,y) co-ordinate
  - Fixation duration : time duration
- Difficult to visualize EM data if (i) Targets move on the display, (ii) Overlap with each other, (iii) Change their shape with time

Objective

- Develop framework for analysis & visualization of EM data :
  - To show the **temporal dynamics** of EM data
  - Develop **metrics for quantifying** the target importance
  - Dynamics of **target importance** over time

Methodology

Step 1 EM Data collection → **Step 2** AOI fixation sequence → **Step 3** AOI transition matrix → **Step 4** Visualization

Step 1: EM data collection

- **Scenario**: Simulated air traffic scenario (20 min.) – avg. 24 aircraft per frame, min 7and max 30 aircraft
- **Task**: Routing & avoid conflict between aircraft
  - Min. Vertical separation of 1000 ft, Horizontal separation 5nmi
- **Participants**: Retired Air Traffic Controllers (ATCs)
- **Hardware**: 19.83×19.83 inch monitor, EyeWorks eye tracker
- **Software**: Matlab, R, EyeWorks, I-Sim

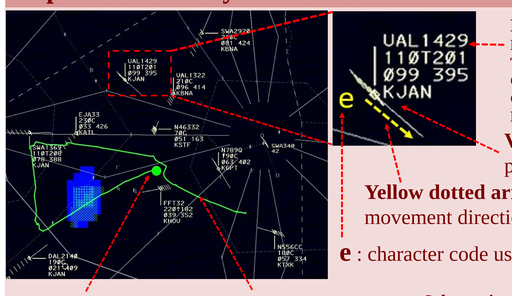
Sample EM data

X pos. (Pixels)	Y pos. (Pixels)	Start time (secs)	End time (secs)
50	40	1	1.5
400	500	1.8	2.2

Target locations data

Time	Target Name	Target corner co-ordinates 1	2	3	4
1	A	(10,15)	(15,15)	(15,10)	(10,10)
1	B	(30,48)	(35,48)	(35,37)	(30,37)
…	…	…	…	…	…
2	A	(12,18)	(12,18)	(12,18)	(12,18)
2	B	(35,53)	(40,53)	(40,42)	(35,42)

Step 2.1 : Create dynamic AOI



Data Tag
Name: United Airline 1429
Target Altitude: 11,000 ft
Current Altitude: 20,100 ft
Ground Speed : 395 knots
Destination: Jackson

Velocity vector : size proportional to speed

Yellow dotted arrow: aircraft movement direction on display

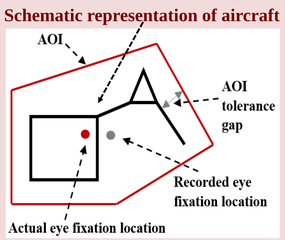
e : character code used for aircraft

Eye fixation

Saccades

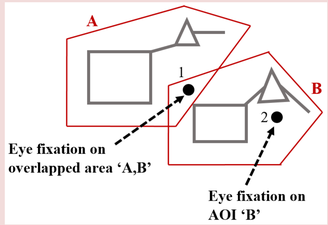
- Difficult to map **eye fixation (EF)** location with the aircraft if:
  - Aircraft change their shape and position
  - **Visual angle accuracy** of Eye tracker
  - **Area of Interest (AOI)** – a convex hull boundary to replace the aircraft

Schematic representation of aircraft



- If EF falls within AOI it is mapped with the respective aircraft
- **AOI gap tolerance** – to account for visual angle accuracy issue
- AOI size is little bigger from the schematic representation

Step 2.2 : Develop AOI fixation sequence



- EF **'1'** falls on both AOI **'A'** and **'B' - 1** mapped with **'A,B'**
- EF **'2'** falls on AOI **'B'- 2** mapped with **'B'**
- Developed AOI fixation sequence – **(A,B)B**

Step 3 : AOI transition matrix

AOI fixation sequence

ABAB(A,C)BC

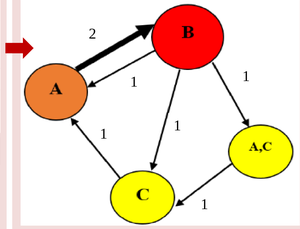
- Eye fixation duration data on AOIs are stored separately

AOI transition matrix

From \ To	A	B	C	A,C
A	-	2	0	0
B	1	-	1	1
C	1	0	-	0
A,C	0	1	0	-
	2	3	1	1

Total number of fixations

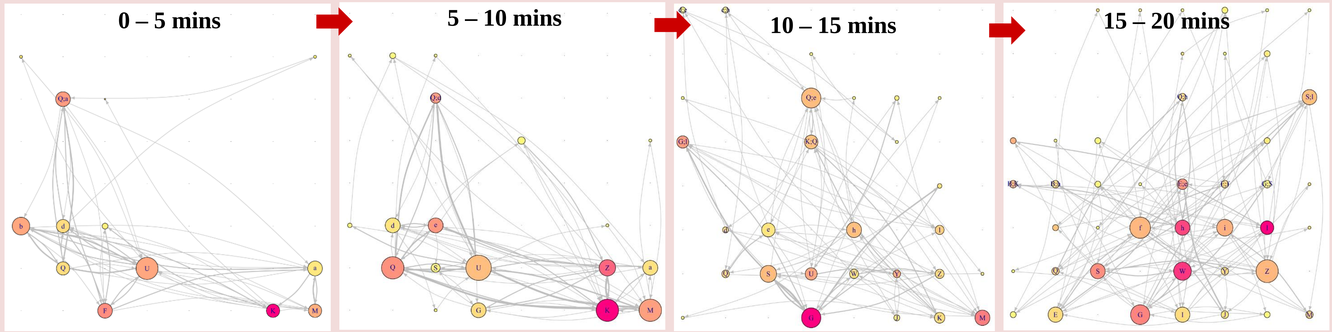
Step 4 :Visualization



- **Network Nodes – AOIs**
- Node size ∝ total # of EF on AOI
- Node colour ∝ EF duration
  - Red colour – high duration
  - Yellow colour – low duration
- **Network Edge – AOI transition**
- Edge thickness ∝ # of transitions
- # of fixation ∝ AOI importance
- Fixation duration ∝ AOI importance

Results

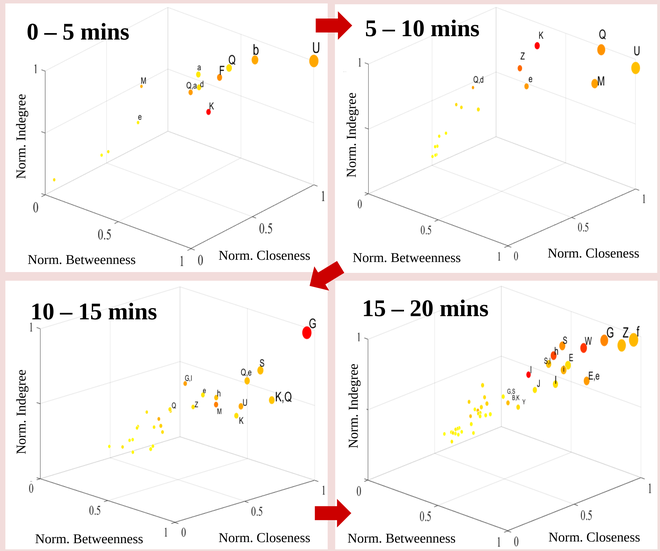
Network visualization of eye movement data



Pros -
- Easy to show important AOIs, AOI pairs having high transition for various time segments **(TS)**
- Grid layout helps to show the change in target importance over various TS
  - 'K' importance- grows from 0-5 to 5-10, reduces in 10-15 & vanishes in 15-20

Cons -
- Only 1 TS per graph
- 1 participant per plot

Norm cube plot -

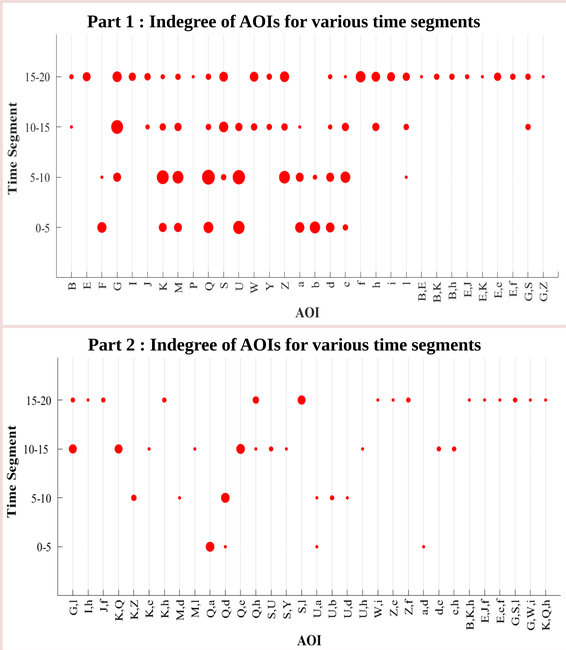


Uses normalized value of 3 centrality metrics
- **Indegree** - # of incoming transition to the AOI
- **Closeness** – Avg. distance from other AOIs in network
- **Betweenness** – fraction of shortest connecting paths passing through the AOI

Pros -
- 3 metrics in 1 plot

Cons –
- 1 TS per plot
- 1 participant per plot

Dot plot -



- Size of the dot ∝ indegree value of the AOI
- Other metrics can also be used
- Absence of dot shows the AOI was absent during that TM

Pros -
- Shows the dynamics of metric value change for all time segments for all AOIs
- Easy to identify the impt. AOIs for any TS
- All TS in a single plot

Cons –
- 1 metric per plot
- 1 participant per plot

Conclusion

- Developed dynamic AOI method to map EM data with dynamic objects
- Developed network based approach to visualize EM data
- Application of 3 centrality metrics to analyze and visualize the scanning strategy
- Developed three different visualization schemes which caters to different visualization requirements
- Require a mapping between the developed plots and the video recording of the EM data to understand the context behind why some targets were more important

Limitations & Future work
- Absence of multiparticipant data analysis in current methods
- Visualization scheme to aid comparison across multiparticipant EM data

Reference

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